WW-F-406D
October 6, 1980
SUPERSEDING
WW-F-406C
10 October 1974

FEDERAL SPECIFICATION

FLANGES, CAST-IRON (CLASSES 125 AND 250) AND BRONZE (CLASSES 150 AND 300)

This specification was approved by the Commissioner, Federal Supply Services, General Services Administration, for use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers cast-iron and bronze flanges for general application.

1.2 Classification. Flanges shall be of the following materials, classes, and types as specified (see 1.2.2 and 6.2).

1.2.1 Materials, types, and classes.

Material.

A - Cast iron.
B - Bronze.

Class.

C - 125.
D - 150.
E - 250.
F - 300.

Type.

C - Companion.
B - Blind.
R - Reducing threaded.

Size.

1/2 through 48 inch nominal in accordance with table I.

FSC 4730

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TABLE I. Code number to type and size of flange.

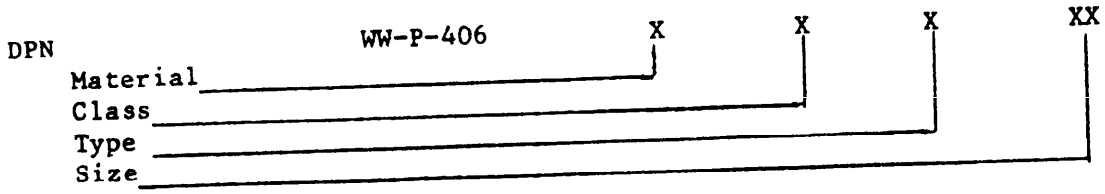
Type of Flange			Nominal pipe size	O.D. of reducing flange
Threaded companion	Blind	Threaded reducing		
01	25		1/2	
02	26		3/4	
03	27		1	
		49	1	5
		50	1	6
04	28		1-1/4	
		51	1-1/4	6
05	29		1-1/2	
		52	1-1/2	6
		53	1-1/2	7
		54	1-1/2	7-1/2
		55	1-1/2	9
06	30		2	
		56	2	7
		57	2	7-1/2
		58	2	8-1/4
		59	2	9
		60	2	11
07	31		2-1/2	
		61	2-1/2	7-1/2
		62	2-1/2	8-1/4
		63	2-1/2	9
		64	2-1/2	11
08	32		3	
		65	3	8-1/2
		66	3	9
		67	3	10
		68	3	11
		69	3	13-1/2
09	33		3-1/2	
		70	3-1/2	9
10	34		4	
		71	4	10
		72	4	11
		73	4	12-1/2
		74	4	13-1/2
11	35		5	
		75	5	11
		76	5	12-1/2
		77	5	13-1/2
12	36		6	
		78	6	13-1/2
		79	6	15
		80	6	16
13	37		8	
		81	8	16
		82	8	19
14	38		10	
		83	10	19

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TABLE I. Code number to type and size of flange continued.

Type of Flange			Nominal pipe size	O.D. of reducing flange
Threaded companion	Blind	Threaded reducing		
15	39		12	
16	40		14	
17	41		16	
18	42		18	
19	43		20	
20	44		24	
21	45		30	
22	46		36	
23	47		42	
24	48		48	

1.2.2 Definitive Part Number System (DPN). Flanges covered by this specification shall be identified by a DPN. This part number is intended for cataloging and ordering purposes only. The DPN shall consist of the Federal Specification number plus five letters or digits (see 1.2.1). The DPN shall be written as follows:



2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

(Activities outside the Federal Government may obtain copies of Federal Specifications, standards, and commercial item descriptions as outlined under General Information in the Index of Federal Specifications, Standards and Commercial Item Descriptions. The Index, which includes cumulative bimonthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification, other Federal specifications and commercial item descriptions required by activities outside the Federal Government for bidding purposes are available without charge from General Services Administration Business Service Centers in Boston; New York; Philadelphia; Washington, DC; Atlanta; Chicago; Kansas City, MO; Fort Worth; Houston; Denver; San Francisco; Los Angeles; and Seattle, WA.

(Federal Government activities may obtain copies of Federal specifications, standards, and commercial item descriptions and the Index of Federal Specification, Standards and Commercial Item Descriptions from established distribution points in their agencies.)

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Military Specification:

MIL-V-3 - Valves, Fittings, and Flanges (except for systems indicated herein): Packaging of.

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inpection by Attributes.
MIL-STD-147 - Palletized and Containerized Unit Loads 40 inch by 48 inch Pallet Skids, Runners or Pallet-Type Base.

(Copies of Military specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

American National Standards Institute (ANSI) Standards:

B16.1 - Cast Iron Pipe Flanges and Flanged Fittings 25, 125, 250, and 800 Pound (lb).
B16.21 - Nonmetallic Gaskets for Pipe Flanges.
B16.24 - Bronze Flanges and Flanged Fittings, 150, and 300 lb.

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018.)

American Society for Testing and Materials (ASTM) Standards:

A126 - Standard Specification for Gray Iron Castings for Valves Flanges and Pipe Fittings.
A307 - Standard Specification for Carbon Steel Externally Threaded Standard Fasteners.
B61 - Standard Specification for Steam or Valve Bronze Castings.
B62 - Standard Specification for Composition Bronze or Ounce Metal Castings.
E54 - Standard Method for Chemical Analysis of Special Brasses and Bronzes.
E59 - Sampling of Steel and Iron for Determining Chemical Composition.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

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Manufacturers Standardization Society of the Valve and Fittings Industry:

SP-6 - Standard Finishes for Contact Faces of Pipe Flanges and and Connecting-End Flanges of Valves and Fittings.

(Application for copies should be addressed to the Manufacturers Standardization Society of the Valve and Fittings Industry, 1815 North Fort Myer Drive, Arlington, VA 22209.)

3. REQUIREMENTS

3.1 Materials. Materials used shall be free from defects which would adversely affect the performance or maintainability of individual components or of the overall assembly. Materials not specified herein shall be of the same quality used for the intended purpose in commercial practice. Unless otherwise specified herein, all material incorporated in the work covered by this specification are to be new and fabricated using materials produced from recovered materials to the maximum extent possible without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products are allowed under this specification unless otherwise specifically specified.

3.2 Design.

3.2.1 General. Flanges shall conform to all requirements of the following ANSI standards, as applicable:

Classes 125 and 250 - B16.1.
Classes 150 and 300 - B16.24.

3.2.2 Reducing and blind flanges. The sizes of reducing flanges or blind flanges shall be designated in terms of the nominal pipe size and the flange outside diameter.

3.2.3 Facings. Unless otherwise specified (see 6.2), the faces of bronze flanges shall be provided with two concentric gasket-retaining grooves not exceeding the maximum diameters specified in ANSI B16.24. These grooves shall not exceed a depth of 1/16 inch or a width of 1/16 inch. Flanges should be flat faced finished smooth or serrated in accordance with MSS SP-6.

3.2.4 Sizes. The sizes shall be as specified in 6.2, in accordance with the appropriate ANSI standards.

3.3 Markings. In addition to the manufacturer's name or trademark, the flanges shall bear numerals indicating the applicable pressure class and nominal pipe size.

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3.4 Bolts and nuts. When specified (see 6.2), the required number of bolts and nuts conforming to ASTM A307 and the applicable ANSI standard (see 3.1), shall be furnished with each flange.

3.5 Gaskets. When specified (see 6.2), a gasket in accordance with ANSI B16.21 shall be provided with each flange. This gasket shall conform to the requirements of ANSI B16.1 or B16.24, as applicable.

3.6 Workmanship. The flanges shall be clean, free from scale, smooth, and of proper dimensions. The flanges shall be free from injurious grooving, indentations, flaws, and slivers which would interfere with use of the flange in the application for which it is intended.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Inspection lot. All flanges of the same material, class, type, and size offered for examination at one time shall be considered a lot for purposes of inspection.

4.3 Sampling.

4.3.1 Sampling for examination. A random sample of flanges shall be selected for examination from each lot in accordance with MIL-STD-105 at inspection level II. The Acceptable Quality Level (AQL) shall be 2.5 percent defective for major defects and 4.0 percent for minor defects.

4.4 Examination. Each sample selected in accordance with 4.3.1 shall be inspected to verify compliance with this specification. Examination shall be conducted as specified in Table II.

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TABLE II. Classification of defects.

Categories	Defects
Major defects:	
101	Type, class, and size not as specified.
102	Material not as required; not sound, not free from cold shuts or injurious blowholes, cracks excessive porosity of inclusions.
103	Flange not smoothly cored or true to form, not thoroughly cleaned of sand, scale, rust or corrosion, and all irregularities.
104	Flange shape not in accordance with specified dimensions.
105	Machining incomplete, surfaces not smooth, not square and flat (as applicable), evidence of burrs, gouges, tool marks, sharp edges, or areas failing to "clean-up" in final machining.
106	Internal diameter of thread, pitch circle, and flange outside diameter not concentric to each other.
107	Flange and center boss, outside diameters, and thicknesses, number and diameter of bolt holes, diameter of bolt circle, chordal dimension between bolt holes, and thickness of bosses not within requirements.
109	Marking not as specified.
Minor defects:	
201	Fillets and radii missing, incomplete, or not well-rounded.

4.5 Chemical analysis. Samples selected for chemical analysis, when specified in 4.3.2, shall be tested in accordance with ASTM E54 or E59, as applicable, to determine conformance with the requirements specified in 3.2.1.

4.6 Inspection of preparation for delivery. The preservation packaging, packing, and marking of the flanges shall be inspected to determine conformance with the applicable requirements of section 5.

5. PREPARATION FOR DELIVERY

5.1 Preservation packaging, packing, and marking. Unless otherwise specified (see 6.2), the flanges shall be preserved, packaged, packed, and marked in accordance with the contractors standard practice and as specified in the contract. When specified (see 6.2), the flanges shall be preserved, packaged, packed and marked in accordance with MIL-V-3 with the level of protection specified.

6. NOTES

6.1 Intended use. Flanges covered herein are intended for general application.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Material, class, and type required (see 1.2).

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(c) Plain facing for bronze flanges, if required (see 3.2.3).
(d) Sizes required see (3.2.4).
(e) Bolts and nuts, if required (see 3.4).
(f) Gaskets, if required (see 3.5).
(g) When chemical analysis is required (see 4.3.2).
(h) When preservation packaging, packing, and marking shall be in accordance with MIL-V-3 and level of protection required (see 5.1).

MILITARY CUSTODIANS:

Army - ME
Navy - YD
Air Force - 99

User activities:

Navy - CG, MC

CIVIL AGENCY COORDINATING ACTIVITIES:

GSA - FSS
HEW - FED
INT - BPA

Preparing activity:

Navy - YD

Project No. 4730-0572

Orders for this publication are to be placed with General Services Administration, acting as an agent for the Superintendent of Documents. See section 2 of this specification to obtain extra copies and other documents referenced herein.

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